Xiangtian Shi

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RESEARCH INTERESTS

Physics-based modeling and simulation, optimal control, physics-informed learning, and sim-to-real robotics.

EDUCATION

ETH Zürich, Switzerland 2026 – Present
M.Sc. in Robotics, Systems and Control

KTH Royal Institute of Technology, Sweden 2023 – 2026
B.Sc. in Vehicle Engineering (GPA: 4.9/5.0)

RESEARCH EXPERIENCE

Biomimetic Flapping-Wing Robot Sim-to-Real Research Jun 2026 – Present
Visiting Student Researcher, Tao Du's Group, Tsinghua IIIS, Tsinghua University, China

- Developing a wireless control interface for a biomimetic flapping-wing robot using embedded systems and SBUS communication.
- Designing mechanical components for an experimental data-acquisition setup supporting sim-to-real validation.
- Contributing to the simulation workflow used to connect model development with physical experiments.

PINNs for Optimal Trajectory Planning of Unicycle Robots Jan 2026 – Jun 2026
Bachelor's Thesis, KTH Royal Institute of Technology, Sweden

- Developed PyTorch frameworks for physics-informed trajectory optimization under kinematic and control constraints.
- Evaluated loss balancing, physics-based constraint enforcement, and obstacle-avoidance strategies.

SELECTED TECHNICAL PROJECTS

Rocket Avionics Engineer Jan 2025 – Sep 2025
AESIR Student Rocketry Association, KTH Royal Institute of Technology, Sweden

- Reduced apogee-prediction error from 10% to 1.3% in RocketPy reference simulations.
- Designed and tuned a PID controller for active stabilization, achieving altitude control within 2.7% error at 3,000 m.

Solar-Powered Model Vehicle Jan 2024 – Jun 2024
Project Course, KTH Royal Institute of Technology, Sweden

- Designed and built a model vehicle integrating solar power, mechanical components, and electrical subsystems.
- Implemented Arduino-based control and evaluated the integrated system through performance testing.

TECHNICAL SKILLS

Programming: Python, MATLAB, Arduino/C++

Tools: PyTorch, Git, CAD, ANSYS, RocketPy

Methods: Optimization, numerical methods, system modeling, machine learning

LANGUAGES

Norwegian (Native)
English (Fluent; IELTS Academic 7.5)
Chinese (Native)
Swedish (Fluent; B2)

WORK EXPERIENCE

Teaching Assistant in Electrical Engineering Oct 2025 – Jun 2026
Unit of Mechatronics, KTH Royal Institute of Technology, Sweden

- Supported laboratory instruction in circuit theory, signal processing, and mechatronics.
- Evaluated assignments and assisted with course coordination.

Mathematics Tutor May 2023 – Jan 2025
MentorNorge AS, Norway

- Delivered individualized mathematics tutoring adapted to each student's learning needs.

Kindergarten Assistant Oct 2022 – Jun 2023
Høvik kindergarten, Norway

- Supported children aged 2–4 in daily routines and structured creative and educational activities.

EXTRACURRICULAR ACTIVITIES

KTH Student Council, International Group Sep 2023 – Jun 2024
KTH Royal Institute of Technology, Sweden

- Organized social and cultural events supporting integration among international students.

Competitive Esports 2019 – 2022
PUBG Mobile team competition

- Won the PUBG Mobile Pro League Western Europe Spring 2021 in a 16-team professional field and competed internationally.

HOBBIES AND INTERESTS

Violin: 10+ years of practice; regional competition winner

Table tennis: Club member; participated in regional competitions

Outdoor activities: Running (Stockholm Half Marathon 2025), hiking, and skiing